

High temperature superconductor current leads for fusion machines

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ABSTRACT

All large fusion experimental machines presently under construction like the tokamak ITER, the stellarator Wendelstein 7-X (W7-X), and the satellite tokamak JT-60SA will use high temperature superconductor (HTS) current leads to reduce the cryogenic load and operational cost. The Karlsruhe Institute of Technology (KIT which is a merger of former Forschungszentrum Karlsruhe and University Karlsruhe) designed, constructed and successfully tested a 68 kA HTS current lead demonstrator for ITER which led to the decision by ITER IO to use HTS current leads for the ITER magnet system, and the Chinese Domestic Agency is in charge of providing these leads as a part of the superconducting feeder system. KIT is in charge of design, construction and test of HTS current leads for W7-X and JT-60SA. The paper discusses the status of the current lead development and describes the results of the W7-X prototype test which has been performed recently.

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1. Introduction

Since the discovery of high temperature superconductivity (HTS) in 1986 there has been a tremendous progress in R&D of HTS material, wires and applications. Especially for current lead application, HTS offers considerable economic benefits because the external field is usually well below 0.5 T allowing an operating temperature not much lower than the critical temperature of the HTS material. Superconducting magnets cooled by 4.5 K super-critical helium require a metallic connection to the power supplies located at room temperature. Using HTS material the cooling power consumption needed for a HTS current lead (HTS-CL) is reduced by a factor of 3–4.

Already in the early 1990s and parallel to the detailed CERN activities for the Large Hadron Collider (LHC) [1], KIT and CRPP started a development program for a HTS-CL in the frame of the EU fusion technology program which ended with the construction and successful testing of the 68 kA ITER HTS current lead demonstrator [2–10]. The development and successful test of the 68 kA ITER HTS-CL demonstrator has brought ITER to the decision to use HTS-CL for the ITER magnet system which can save about 22 kW cold refrigerator power.

Other fusion devices in operation or under construction like EAST (China), W7-X (Germany) and JT-60SA (Japan) use or plan to use HTS-CL. Academia Sinica Institute for Plasma Physics (ASIPP)

has constructed liquid nitrogen (LN₂) cooled HTS-CL for the EAST tokamak [12]. KIT is responsible for the design, construction and testing of the HTS-CLs for the stellarator W7-X [13] and the satellite tokamak JT-60SA [14].

2. Design guide lines for HTS current leads

The general design of HTS-CL follows some basic principles which are determined by the electromagnetic properties of the HTS material. All leads for high current capacities in operation or under construction use Bi-2223/AgAu tapes as basic component. They consist of a HTS part covering the temperature range between 4.5 K and an intermediate temperature $T_{100\%HTS}$ – mostly at about 60–65 K – and a conventional heat exchanger (HEX) in the range of $T_{100\%HTS}$ to room temperature. The HTS part is mostly cooled by heat conduction from the 4.5 K end whereas the heat exchanger is cooled with helium at approx. 50 K.

Due to the strong and anisotropic dependence of the critical current of Bi-2223 as a function of magnetic field, the HTS part has a cylindrical shape where on its outer surface the HTS tapes are located such to minimize the magnetic field perpendicular to the broad face of the tapes. Fig. 1 (left) shows the dependence of the normalized I_c as a function of the applied magnetic field for different temperatures. Fig. 1 (right) shows a schematic of the cylindrical design of the HTS.

The copper HEX is made with different designs (perforated plate, fin-type, foil type). In principle the efficiency of all HEX types are similar and the final choice is made for manufacturing reasons.

Critical design parameters which influence strongly the performance are:

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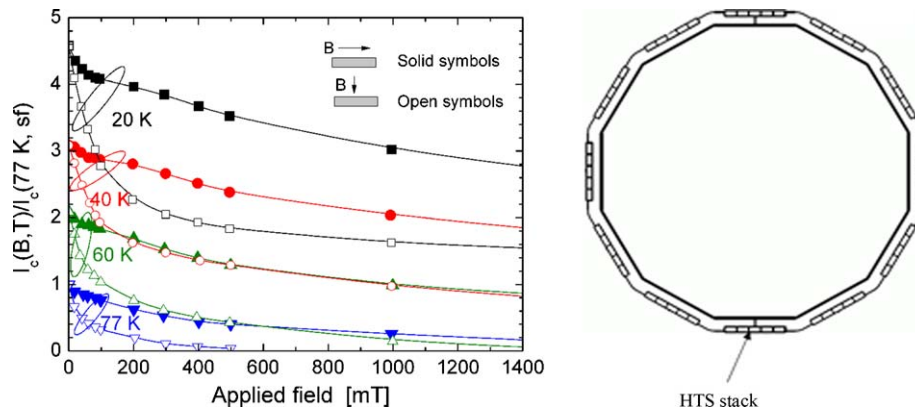


Fig. 1. Left: Dependence of the normalized I_c as a function of the applied magnetic field for different temperatures. Right: Schematic of the cylindrical design of the HTS part.

- Contact resistances between the HTS elements and the cold end joint and the HEX respectively which determines the cold end heat load as well as the 50 K He mass flow rate at operation current.
- Efficiency of the HEX which determines the 50 K He mass flow rate and the length of the HEX.
- Temperature margin of the HTS elements which is related to the current sharing temperature at operation conditions. It determines the performance of the current lead in case of a loss of 50 K He flow (LOFA) and quench.
- Paschen-tight high voltage insulation with integrated mechanical connection to the cryostat. To simplify the electrical insulation and to provide space for the instrumentation wiring routed to the room temperature end and avoiding additional HV feedthroughs at the cryostat side, a separated insulation vacuum around the current lead is mandatory. In addition, a bellow at the room temperature side of the vacuum tube is required to compensate for differential shrinkage between the stainless steel tube and the copper heat exchanger during cool down.

In the following examples of HTS-CL for fusion will be described and test results will be briefly summarized.

3. HTS current leads for ITER

3.1. EU development program towards ITER

A 68 kA HTS-CL demonstrator for ITER was developed and built by KIT, Germany, in collaboration with CRPP, Switzerland in the frame of an EU Fusion Technology task. The maximum steady state current of the HTS-CL was 80 kA which exceeds the value of 68 kA needed for the ITER TF coils. In addition it was shown that even when the 50 K He-cooling has been blocked, a current of 68 kA could be carried for more than 6 min which shows the stability of this current lead. It could be shown that the He-refrigerator power consumption was reduced by a factor of 4 compared to conventional current leads. Extrapolation for the ITER magnet system resulted in saving about 22 kW cold refrigerator power at 4.5 K. During the different tests, the HTS-CL was cooled with He at an inlet temperature of 50 K, 80 K and also with LN_2 at atmospheric and subatmospheric pressure. A picture of the 68 kA ITER HTS-CL demonstrator is shown in Fig. 2. Details can be found in [8–11].

3.2. ITER program

ITER has a total of 60 single current leads, 18 for the TF coils, 12 for the CS coils, 12 for the PF coils and 18 for the CC correction

coils, with a total nominal operating current capacity of approximately 2.6 MA. As part of the in-kind contribution to ITER, China is currently constructing the low temperature feeder system which includes the current leads. After a design change made by ITER IO, these current leads will be HTS current leads. Table 1 shows the ITER design specification [15].

China DA has manufactured and tested trial current leads for 68 kA, 52 kA and 10 kA [16–18]. All leads are of the binary type: they consist of a HTS part covering the temperature range between 4.5 K and 65 K and a conventional heat exchanger in the range of 65 K to room temperature. Two different heat exchanger designs are being considered: fin-type (68 kA) and foil-type (52 kA, 10 kA). The HTS part is cooled by heat conduction from the 4.5 K end whereas the heat exchanger is cooled with 50 K helium. The HTS material used in the current leads is Bi-2223/AgAu tape made by various manufacturers. Fig. 3 shows a picture of the 68 kA HTS trial current lead [17].

4. Current leads for W7-X and JT-60SA

The stellarator W7-X presently under construction at the Greifswald branch of the Max-Planck-Institut für Plasmaphysik consists of 50 non-planar and 20 planar coils with a maximum conductor current of 17.6 kA. In total 14 current leads are required (maximum design current $I_{max} = 18.2$ kA, nom. current $I_{nom} = 14$ kA) [13].

Main requirements of the W7-X machine dominate the design of the current leads:

- Mounting in upside-down position, i.e. the cold end of the current lead is at the top and the warm end at the bottom side.
- The necessity of using low-Co stainless steel material and the limitation of the amount of silver to reduce activation by neutrons.
- The location of the current leads very close to the magnets results in a rather high magnetic stray field.
- Paschen tightness and test voltage of 13 kV of the current lead and its instrumentation has to be assured.

In the frame of the Broader Approach Agreement between Japan and EU and concomitantly to the ITER project, the satellite tokamak project JT-60SA has been agreed. The magnet system consists of 18 TF coils (25.7 kA), 4 CS modules (20 kA) and 7 PF coils (20 kA) [14].

Following the commitment of the German Government to the EU and described in the bilateral procurement arrangement between Japan and EU and the agreement of understanding between EU and KIT, KIT is responsible for design, constructing and testing the current leads. In total 6 leads for a maximum current of 26 kA and 20 leads with a maximum current of 20 kA, mounted

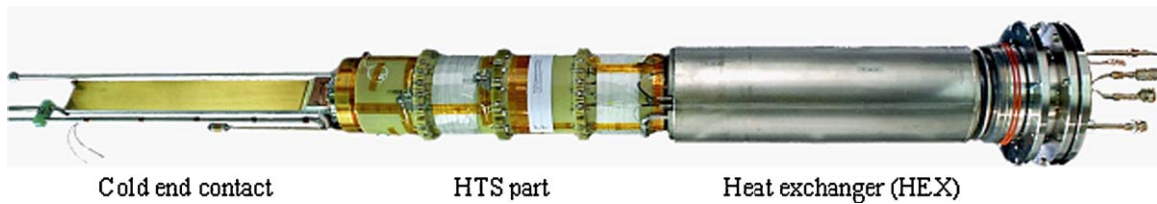


Fig. 2. 68 kA HTS current lead developed at KIT using Bi-2223/AgAu tapes.

Table 1

Main specifications of the HTS current leads for ITER, W7-X and JT-60SA.

Parameter	ITER				W7-X	JT-60SA	
	TF	CS	PF	CC		TF	CS/EF
Number of current lead pairs	9	6	5	9	7	3	3/7
Maximum design current (kA)	70	46	57	10	18.2	25.7	20
Test voltage to ground (kV)	±19	±29	±25	±7	±13	±3.8	±21
Peak external field $B_{\perp}$ (T)			0.03		0.09	0.033	
HTS burnout time ^a (s)	15	11.5	18	18	3.5	N/A	N/A
LOFA time (s)	400 ^b	250 ^b	250 ^b	120 ^b	360 ^b	180 ^b	180 ^b
Pressure drop in HEX (MPa)			<0.1		<0.1	<0.4	
Number of thermal cycles			100		100	<200	
Orientation			Horizontal		Cold end at top	Vertical	

^a Burnout time = time at full current after a quench to reach hot spot temperature.

^b LOFA time = time at full current after a LOFA accident in the 50 K He circuit to reach the current sharing temperature in HTS.

in vertical, upright position are required. Since the current leads for JT-60SA are located outside the biological shield, no special needs regarding material are required. Table 1 shows the W7-X and JT-60SA specifications for the current leads.

The design of the HTS-CL for W7-X and JT-60SA follows the same principles as that of the 68 kA ITER HTS current lead demonstrator described above. It consists of a high temperature superconductor (HTS) part covering the temperature range between 4.5 K and 60 K and a conventional fin-type HEX in the range of 60 K to room temperature. The HTS part is cooled by heat conduction from the 4.5 K end whereas the heat exchanger is cooled with 50 K helium. The HTS material used in the current leads is Bi-2223/AgAu tape manufactured by Bruker HTS [20]. The copper HEX consists of a central current carrying copper rod and copper fins around it. The design of the HEX follows the design principle used by CERN for the LHC current leads [1] with some minor modifications.

The high-voltage requirement of W7-X and JT-60SA requested a special design of the electrical insulation system. In the KIT design, the electrical insulation is directly performed on the outer surface of the current lead vacuum vessel by a solid insulation and a G10

flange is integrated to serve as a connection part to the vacuum vessel of the cryostat. Fig. 4 shows a picture of the current lead after assembly before installation into the cryostat.

5. Results of the W7-X HTS prototype current lead test

After completion of the prototype current lead manufacturing the two leads were installed in a test cryostat together with a superconducting short circuit bus bar. The connection is done by clamping, using a 100 μ m thick Indium foil between the two Au-plated copper surfaces of the current lead cold end and the joint box of the bus bar. After closing the vacuum space of the current leads the outer surface of the insulation has been covered with a conductive paint.

The purpose of the prototype test was to confirm the design and to determine the operation parameters [21]. Details of the cryogenic system used during the different test campaigns are described in [22]. The test covered measurement of heat load at 4.5 K with zero current, steady state operation at currents up to 20 kA and long time

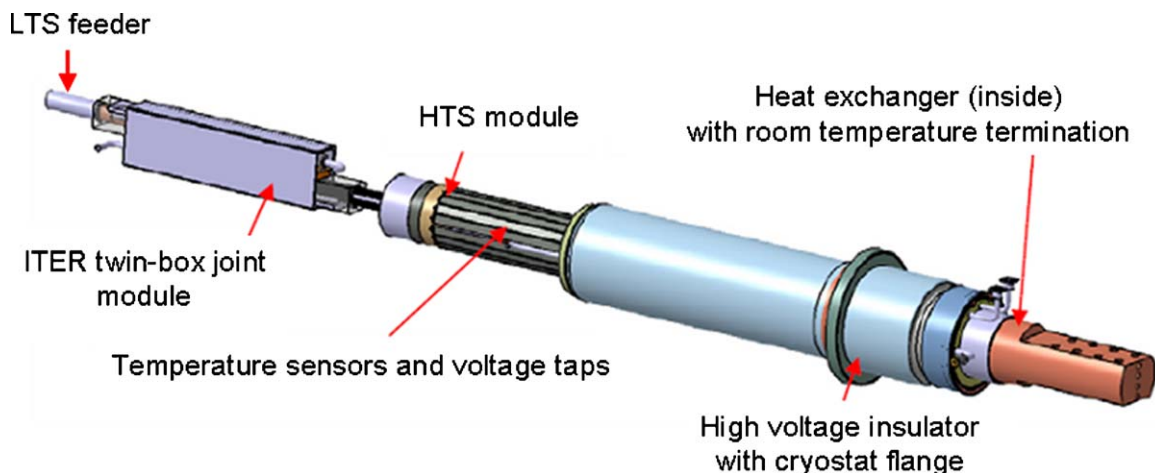


Fig. 3. CAD drawing of the 68 kA HTS trial current lead developed by ASIIPP [19].

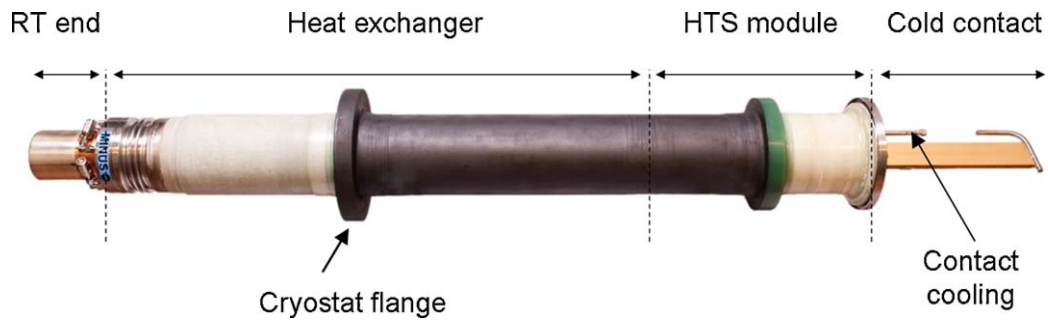


Fig. 4. W7-X prototype HTS current lead.

Table 2

Main results of W7-X prototype current lead tests.

Parameter	Result
Nom./maximum current	14/20 kA
Helium inlet temperature	50 K
Temperature at warm end of HTS	60 K
Cold end resistance (incl. clamp to BB ^a)	(11.5 ± 0.6) nΩ
Warm end resistance HTS (incl. Cu part)	10 nΩ
Voltage drop of HEX at 14/18.2 kA	51.9/80.8 mV
He mass flow rate at 0 kA ^a	0.53 g/s
Heat load at 0 kA ^b	(2.4 ± 1) W
He mass flow rate at 18.2 kA ^b	1.38 g/s
Pressure drop at 18.2 kA ^b	(250 ± 50) mbar
Heat load at 18.2 kA ^b	(6.3 ± 1) W
He mass flow rate at 20 kA	1.58 g/s
Pressure drop at 20 kA	(320 ± 50) mbar
Heat load at 20 kA	(7.0 ± 1) W

^a BB = bus bar.^b Nominal conditions, i.e. $T_{100\%HTS} = 60$ K.

test with 18.2 kA, current ramp test (W7-X scenario), and quench and loss of mass flow (LOFA) tests.

Table 2 summarizes the main results of the prototype current lead tests. It could be demonstrated that the prototype HTS current leads for W7-X behave as expected. The design is validated. The long time stability is excellent and mainly determined by the stability of the cryogenic supply. The stable and optimal operation of the current leads is not affected by the upside-down orientation. The heat load @ 4.5 K end in nominal conditions is 2.4 W and the He mass flow rate at 18.2 kA is 1.38 g/s. The temperature margin has been measured to be >26 K which gives enough safety margins under W7-X conditions (incl. magnetic stray field from the torus). Under LOFA conditions at 18.2 kA, the time until quench is approx. 18 min. In case of a linear ramp down of the magnet current the maximum

temperature of the HTS-module was 72.8 K which is low enough to prevent the HTS from quenching even under W7-X conditions. Fig. 5 shows the specific helium mass flow rate as a function of the normalized current for the W7-X HEX and a comparison to the HEX of the ITER demonstrator and the ITER trial leads.

6. Conclusions

KIT has a long time experience in the development of HTS current leads for fusion machines. As the completion of a 8 years development program the 68 kA ITER HTS current demonstrator was constructed and successfully tested. Since the ITER feeder package has been overtaken by the Chinese Home Team, the HTS current leads for ITER are developed by CN.

KIT will deliver HTS current leads for the stellarator W7-X and the satellite tokamak JT-60SA. After the development and qualification phase ending with prototype construction, the prototype test was successfully carried out recently.

Nowadays the use of high temperature superconductor material in large current capacity current leads for fusion machines is common practice. The higher investment costs of the current lead manufacturing can be over-compensated by the drastic reduction of investment costs for the refrigerator and by the significantly lower operation costs.

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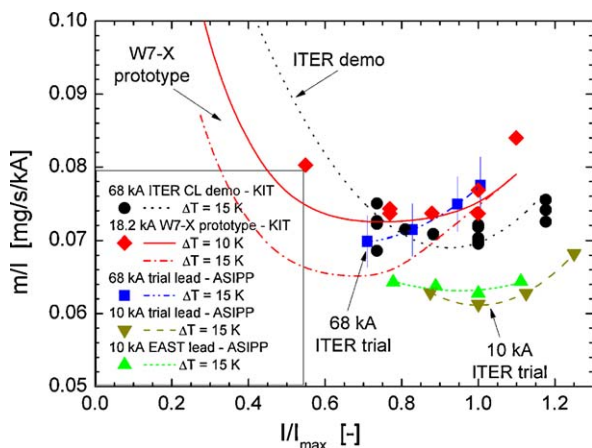


Fig. 5. Specific He mass flow rate vs. normalized current for ITER and W7-X current leads.

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